

neurons when one of them is depolarized (or hyperpolarized) decrease and are therefore smaller in the postsynaptic neuron than in the presynaptic neuron. Under most circumstances, the communication is bidirectional; however, so-called rectifying synapses limit current flow in one direction, as is typical in chemical synapses.

Electrical synapses are useful when information must be conducted rapidly, such as in the escape reflex of some invertebrates. Groups of neurons with these synapses can activate muscles quickly to get the animal out of harm's way. For example, the well-known tail flip reflex of crayfishes involves powerful rectifying electrical synapses. Electrical synapses are also useful when groups of neurons should operate synchronously, as with some hypothalamic neurosecretory neurons. Electrical synapses also have some limitations: They are much less plastic than chemical synapses, and they cannot amplify a signal (whereas an action potential that triggers a chemical synapse could cause a large release of neurotransmitter, thus amplifying the signal).

TAKE-HOME MESSAGES

- Synapses are the locations where one neuron can transfer information to another neuron or a specialized nonneuronal cell. They are found on dendrites and at axon terminals but can also be found on the neuronal cell body.
 - Chemical transmission results in the release of neurotransmitters from the presynaptic neuron and the binding of those neurotransmitters on the postsynaptic neuron, which in turn causes excitatory or inhibitory postsynaptic potentials (EPSPs or IPSPs), depending on the properties of the postsynaptic receptor.
 - Neurotransmitters must be removed from the receptor after binding. This removal can be accomplished by active reuptake back into the presynaptic terminal, enzymatic breakdown of the transmitter in the synaptic cleft, or diffusion of the neurotransmitter away from the region of the synapse.
 - Electrical synapses are different from chemical synapses because they operate by passing current directly from one neuron (presynaptic) to another neuron (postsynaptic) via specialized channels in gap junctions that connect the cytoplasm of one cell directly to the other.
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2.3 Overview of Nervous System Structure

Until now, we have been talking about only one or two neurons at a time. This approach is useful in understanding how neurons transmit information, but it is only one part

of how the nervous system and the brain function. Neural communication depends on patterns of connectivity in the nervous system, the neural “highways” along which information travels from one place to another. Identifying patterns of connectivity in the nervous system is tricky because most neurons are not wired together in simple, serial circuits. Instead, neurons are extensively connected in both serial and parallel circuits.

A single cortical neuron is likely to be innervated by (i.e., receive inputs from) a large number of neurons: A typical cortical neuron has between 1,000 and 5,000 synapses, while a Purkinje neuron in the cerebellum may have up to 200,000 synapses. The axons from these input neurons can originate in widely distributed regions. Thus, there is tremendous convergence in the nervous system, but also divergence, in which a single neuron can project to multiple target neurons in different regions.

Localized interconnected neurons form what is known as a **microcircuit**. They process specific kinds of information and can accomplish sophisticated tasks such as processing sensory information, generating movements, and mediating learning and memory. For example, microcircuits in the retina of the eye process information from small regions of visual space. The patterns of light entering the eye strike photoreceptors, and the information from adjacent photoreceptors is processed through the microcircuitry of the retina to encode spatial information. This encoding permits a much smaller number of retinal ganglion cells to transmit the coded information down the optic nerve.

Although most axons are short projections from neighboring cortical cells, some are quite long, originating in one region of the brain and projecting to another at some distance. For example, the ganglion cells of the retina project via the optic nerve to the lateral geniculate nucleus of the thalamus, and the lateral geniculate neurons project via the optic radiations to primary visual cortex. Axons from neurons in the cortex may reach distant targets by descending below the cortical sheath into the white matter, traveling through long fiber tracts, and then entering another region of cortex, subcortical nucleus, or spinal layer to synapse on another neuron.

These long-distance connections between various brain regions connect to form more complex **neural networks**, which are *macrocircuits* that are made up of multiple embedded microcircuits. Neural networks support more complex analyses, integrating information processing from many microcircuits. Connections between two cortical regions are referred to as *corticocortical* connections, following the convention that the first part of the term identifies the source and the second part identifies the target.

Inputs that originate in subcortical structures such as the thalamus would be referred to as *thalamocortical* connections; the reverse are *corticothalamic* or, more generally, *corticofugal* projections (projections extending from more central structures, like cortex, outward toward the peripheral nervous system).

Ultimately, neural networks are organized into neural systems. So, for example, the microcircuits of the retina, the lateral geniculate nucleus of the thalamus, and the patches of visual cortex are organized to create neural networks such as the thalamocortical network. This progression ultimately leads to the organization of the visual system as a whole, with its numerous cortical areas and subcortical components. (We will discuss the organization of the visual system in detail in Chapter 5.) In the next section we go on a brief anatomical tour of the brain. Early in each of Chapters 4 through 14, a section highlights the anatomy most relevant to the cognitive functions discussed in that chapter. The anatomy presented here and in the coming chapters will help you see which structures of the brain are related to which of its functions.

The two main divisions of the nervous system are the **central nervous system (CNS)**, consisting of the brain and spinal cord, and the **peripheral nervous system (PNS)**, consisting of the nerves (bundles of axons and glial cells) and ganglia (clumps of nerve cell bodies) outside of the CNS (**Figure 2.17**). The CNS can be thought of as the command-and-control center of the nervous system. The PNS represents a courier network that delivers sensory information to the CNS and carries motor commands from the CNS to the muscles. These activities are accomplished through two subsystems: the *somatic motor system* that controls the voluntary muscles of the body, and the *autonomic motor system* that controls the automated visceral functions. Before we concentrate on the CNS, a word about the autonomic nervous system.

The Autonomic Nervous System

The **autonomic nervous system** (also called the *autonomic motor system* or *visceral motor system*) is involved in controlling the involuntary action of smooth muscles, the heart, and various glands. It also has two subdivisions: the *sympathetic* and *parasympathetic* branches (**Figure 2.18**). In general, the sympathetic system uses the neurotransmitter norepinephrine, and the parasympathetic system uses acetylcholine as its transmitter. The two systems frequently operate antagonistically. For example, activation of the sympathetic system increases heart rate, diverts blood from the digestive tract to the somatic

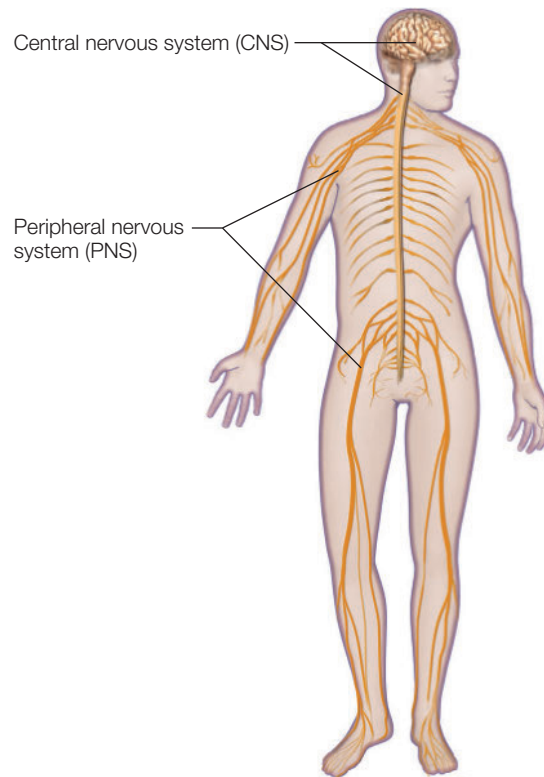


FIGURE 2.17 The peripheral and central nervous systems of the human body.

The nervous system is generally divided into two main parts. The central nervous system (CNS) includes the brain and spinal cord. The peripheral nervous system (PNS), comprising the sensory and motor nerves and associated nerve cell ganglia (groups of neuronal cell bodies), is located outside the central nervous system.

musculature, and prepares the body for action (fight or flight) by stimulating the adrenal glands to release adrenaline. In contrast, activation of the parasympathetic system slows heart rate, stimulates digestion, and in general helps the body with functions germane to maintaining itself (rest and digest).

In the autonomic system, a great deal of specialization takes place that is beyond the scope of this chapter. Still, understanding that the autonomic system is involved in a variety of reflex and involuntary behaviors (mostly below the level of consciousness) is useful for interpreting information presented later in the book. In Chapter 10, on emotion, we will discuss arousal of the autonomic nervous system and how changes in a number of psychophysiological measures tap into emotion-related changes in the autonomic nervous system. For example, changes in skin conductance are related to sweat gland activity, and sweat glands are under the control of the autonomic nervous system.

In the rest of this chapter we focus on the CNS in order to lay the groundwork for the studies of cognition

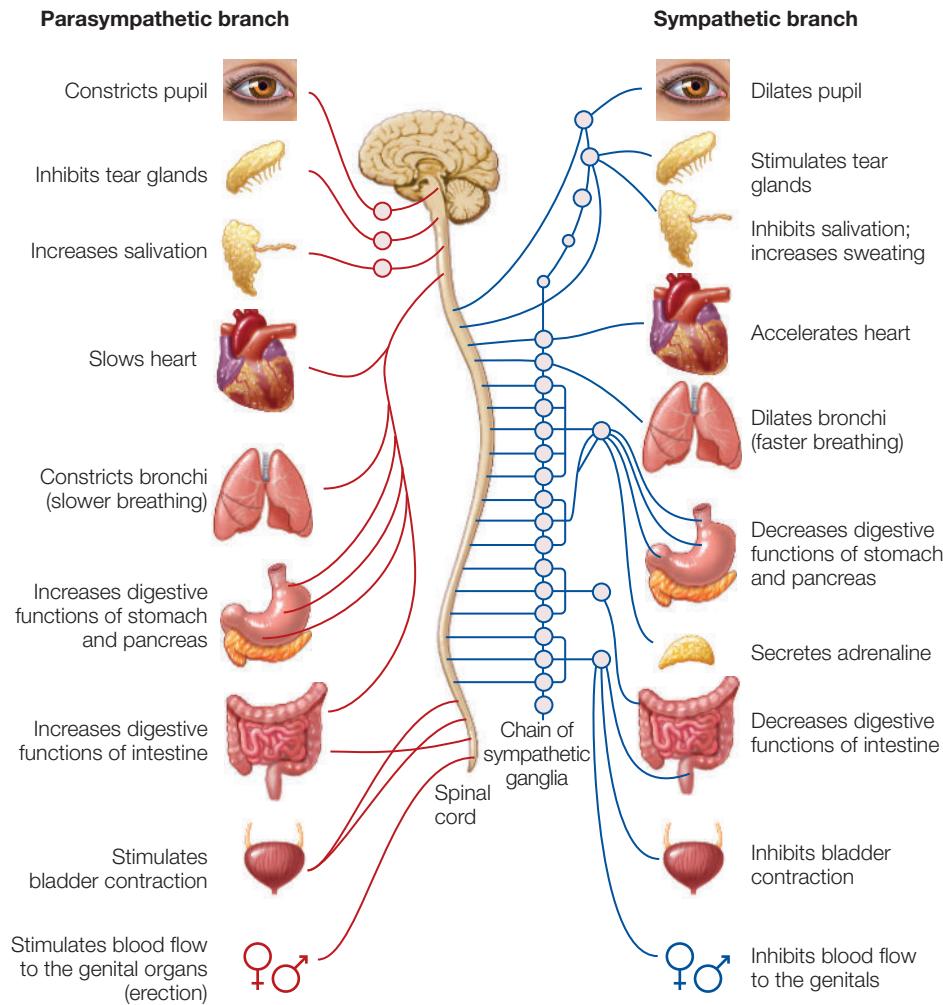


FIGURE 2.18 Organization of the autonomic nervous system.

presented throughout the rest of the book. When we talk about brain anatomy, we will use standard terminology to locate parts of the brain in three-dimensional space (see **Box 2.1**).

The Central Nervous System

The CNS is made up of the brain and spinal cord, and each is covered with three protective membranes, the *meninges*. The outer membrane is the thick *dura mater*; the middle is the *arachnoid mater*; and the inner and most delicate is the *pia mater*, which firmly adheres to the surface of the brain. Between the *arachnoid mater* and the *pia mater* is the *subarachnoid space*, which is filled with *cerebrospinal fluid (CSF)*, as are the brain's ventricles, cisterns, and sulci, along with the central canal of the spinal cord. The brain actually floats in the CSF, which offsets the pressure and damage that would be present if it were merely sitting and scraping on the base of the skull.

CSF also reduces shock to the brain and spinal cord during rapid accelerations or decelerations, such as when we fall, ride roller coasters, or are struck on the head.

Within the brain are four large interconnected cavities called **ventricles** (**Figure 2.19**). The largest are the two lateral ventricles in the cerebrum, which are connected to the more caudal third ventricle in the brain's midline and the fourth ventricle in the brainstem below the cerebellum. The walls of the ventricles contain a system of specialized cells and capillaries, the *choroid plexus*, which produces CSF from blood plasma. The CSF circulates through the ventricles and on to either the subarachnoid space surrounding the brain or the spinal canal. It is reabsorbed in the brain by the *arachnoid villi*, protrusions into the venous system in the sagittal sinus.

In the CNS, neurons are bunched together in various ways (**Figure 2.20**). Two of the most common organizational clusters are the **nucleus** and the **layer**. A nucleus is a relatively compact arrangement of nerve cell bodies

future reward and actual reward (Schultz et al., 1997), and plays a crucial role in motivation and learning. The basal ganglia may also play a big role in reward-based learning and goal-oriented behavior. One summary of basal ganglia function proposes that it combines an organism's sensory and motor context with reward information and passes this integrated information to the motor and prefrontal cortex for a decision (Chakravarthy et al., 2009).

TAKE-HOME MESSAGES

- Many neurochemical systems have nuclei in the brainstem that project widely to the cerebral cortex, limbic system, thalamus, and hypothalamus.
- The cerebellum integrates information about the body and motor commands, and it modifies motor outflow to effect smooth, coordinated movements.
- The thalamus is the relay station for almost all sensory information.
- The hypothalamus is important for the autonomic nervous system and endocrine system. It controls functions necessary for the maintenance of homeostasis. It is also involved in control of the pituitary gland.
- The limbic system includes subcortical and cortical structures that are interconnected and play a role in emotion.
- The basal ganglia are involved in a variety of crucial brain functions, including action selection, action gating, reward-based learning, motor preparation, timing, task switching, and more.

2.5 The Cerebral Cortex

The crowning glory of the cerebrum is its outermost tissue, the cerebral cortex. As described earlier, it is made up of a large sheet of (mostly) layered neurons, crumpled and folded on the surface of the two symmetrical hemispheres like frosting on a cake. The cerebral cortex sits over the top of the core structures that we have been discussing, including parts of the limbic system and basal ganglia, and it surrounds the structures of the diencephalon. The term *cortex* means “bark,” as in tree bark, and in higher mammals and humans it contains many infoldings, or convolutions (**Figure 2.34**). The infoldings of the cortical sheet are called **sulci** (the crevices) and **gyri** (the crowns of the folded tissue that are visible on the surface).

The folds of the human cortex serve two important functions. First, they enable more cortical surface to be packed into the skull. If the human cortex were smoothed out to resemble that of the rat, for example, humans would need to have very large heads. The total surface area of the human cerebral cortex is about 2,200 to 2,400 cm², but because of extensive folding, about two thirds of this area is confined within the depths of the sulci.

Second, having a highly folded cortex brings neurons that are located at some distance from each other along the cortical sheet into closer three-dimensional relationships; for example, the opposing layers of cortex in each gyrus are in closer linear proximity than they would be if the gyri were flattened. Because the axons that make long-distance corticocortical connections run under the

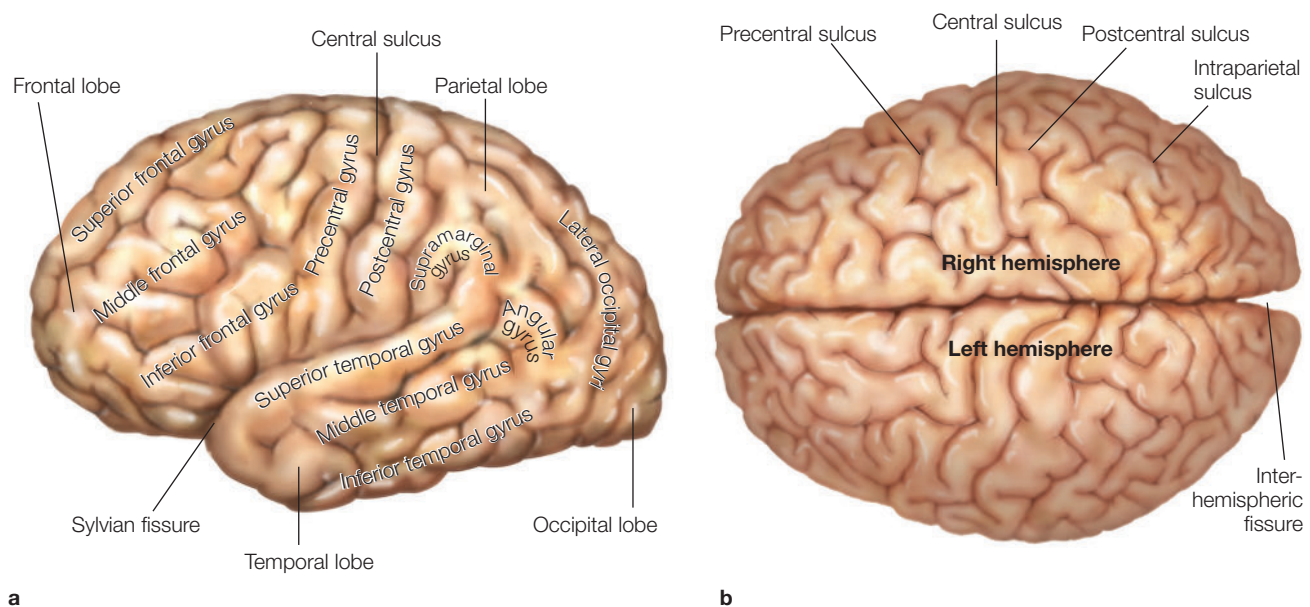


FIGURE 2.34 The cerebral cortex.

Lateral view of the left hemisphere (**a**) and dorsal view of the human brain (**b**). The major features of the cortex include the four cortical lobes and various key gyri. Gyri are separated by sulci and result from the folding of the cerebral cortex that occurs during development of the nervous system.

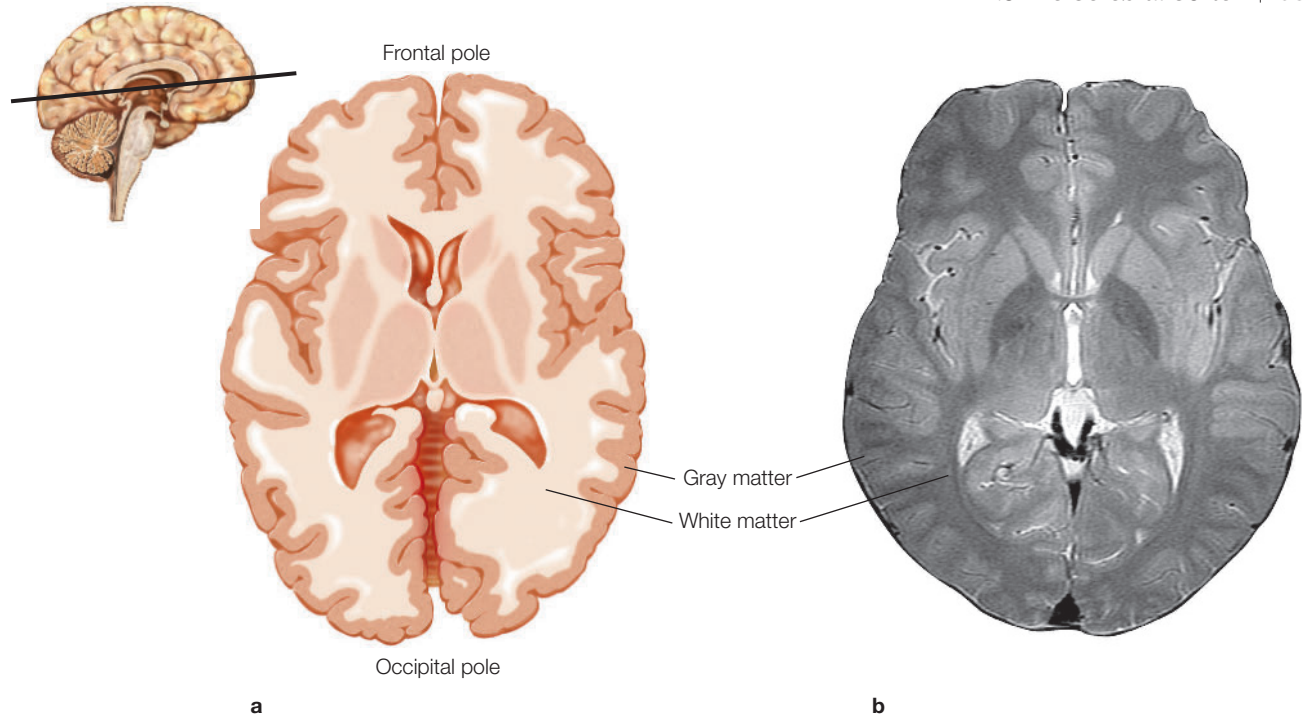


FIGURE 2.35 Cerebral cortex and white matter tracts.

(a) Horizontal section through the cerebral hemispheres at the level indicated at upper left. White matter is composed of myelinated axons, and gray matter is composed primarily of neurons. This diagram shows that the gray matter on the surface of the cerebral hemispheres forms a continuous sheet that is heavily folded. (b) High-resolution structural MRI in a similar plane of section in a living human. This T2 image was obtained on a 4-tesla scanner. Note that the imaging technique used for T2 images makes the white matter appear darker than the gray matter.

cortex through the white matter and do not follow the foldings of the cortical surface in their paths to distant cortical areas, they can project directly to the neurons brought closer together by folding.

The cortex ranges from 1.5 to 4.5 mm in thickness, but in most regions it is approximately 3 mm thick. It contains the cell bodies of neurons, their dendrites, and some of their axons. In addition, the cortex includes axons and axon terminals of neurons projecting to the cortex from other brain regions, such as the subcortical thalamus. The cortex also contains blood vessels.

Because the cerebral cortex has such a high density of cell bodies, it appears grayish in relation to underlying regions that are composed primarily of the axons that connect the neurons of the cerebral cortex to other locations in the brain. These underlying regions appear slightly paler or even white because of their lipid myelin sheaths (Figure 2.35a). As described earlier, this is the reason anatomists coined the terms *gray matter* and *white matter* when referring to areas of cell bodies and axon tracts, respectively. Yet no well-defined anatomical boundaries are visible in the wrinkled cortex. As a result, it can be divided up on the basis of its surface features, its microscopic architecture, or its functional features. We will explore all three of these divisional schemes.

Dividing the Cortex by Surface Features

The cerebral cortex can be divided by the gross anatomical features of the cortical surface. The cortex of each hemisphere has four main divisions, or lobes, that are best seen in a lateral view (Figure 2.36). The **frontal**,

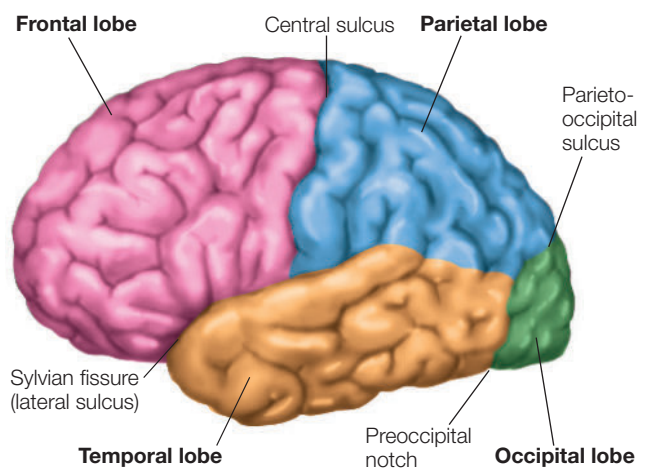


FIGURE 2.36 The four lobes of the cerebral cortex.

This lateral view of the left hemisphere shows the four major lobes of the brain, along with some of the major landmarks that separate them.

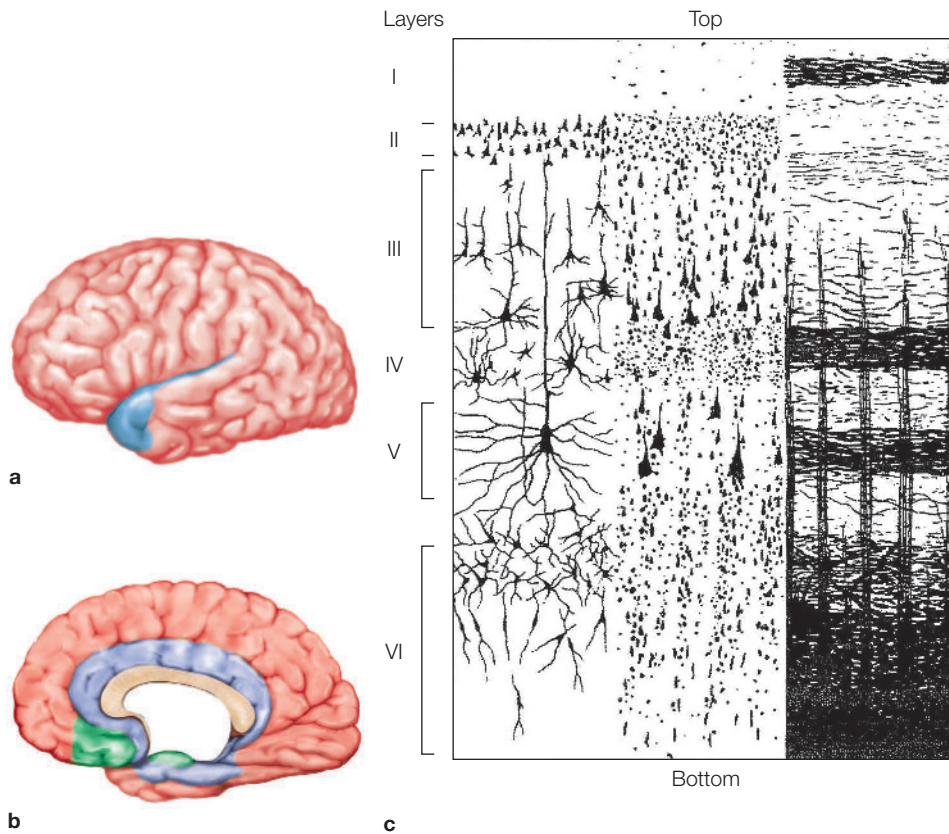


FIGURE 2.38 Cerebral cortex, color-coded to show the regional differences in cortical layering that specify different types of cortex.

(a) Lateral surface of the left hemisphere. (b) Medial surface of the right hemisphere. Neocortex is shown in red, mesocortex in blue, and allocortex in green. (c) Idealized cross section of neocortex, showing a variety of cell types and the patterns produced by three different types of staining techniques. On the left, the Golgi preparation is apparent: Only a few neurons are stained, but each is completely visualized. In the middle, we see primarily cell bodies from the Nissl stain. On the right, we see the fiber tracts in a Weigert stain, which selectively stains myelin.

Dividing the Cortex by Function

The lobes of the cerebral cortex have a variety of functional roles in neural processing. Sometimes we get lucky and can relate the gross anatomical subdivisions of the cerebral cortex to fairly specific functions, as with the precentral gyrus where the primary motor cortex resides, or the face recognition area of the fusiform gyrus. More typically, however, cognitive brain systems are composed of networks whose component parts are located in different lobes of the cortex. In addition, most functions in the brain—whether sensory, motor, or cognitive—rely on both cortical and subcortical components. Nevertheless, the cortex has generally been subdivided into five principal functional subtypes: primary sensory areas, primary motor areas, unimodal association areas, multimodal (heteromodal) association areas, and paralimbic and limbic areas (**Figure 2.39**; reviewed in Mesulam, 2000).

The primary visual, auditory, somatosensory, and motor regions are highly differentiated cytoarchitecturally and also, of course, functionally. A greater portion of the cerebral cortex is given over to what are classically known as association areas that integrate processing. They may be unimodal areas (meaning that they process one type of information—e.g., motor information or visual information) or multimodal (integrating more than one type). The rest of this section provides a beginner's guide to the functional anatomy of the cortex. The detailed functional anatomy of the brain will be revealed in the next twelve chapters. We will start with primary motor and sensory areas and their unimodal association areas.

THE FRONTAL LOBE: MOTOR CORTEX The frontal lobe has two main functional subdivisions: the prefrontal cortex (a higher-order association area that we will discuss later in this section) and the motor cortex

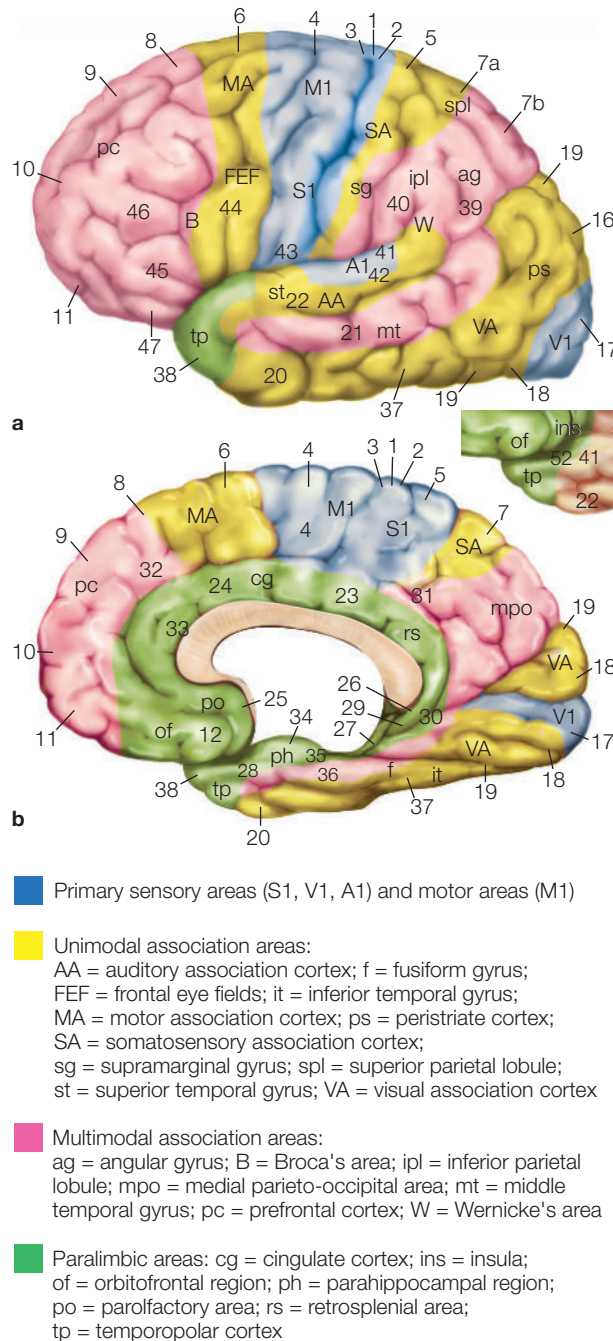


FIGURE 2.39 Functional areas in relationship to Brodmann's map of the human brain.

Unimodal association areas process one type of information. Multimodal association areas integrate more than one type of information. Core limbic structures, such as amygdala, are not shown. (a) Lateral view of the left hemisphere. (b) Midsagittal view of the right hemisphere. The small image shows the insula, not seen in the other views.

(Figure 2.40). The motor cortex sits in front of the central sulcus, beginning in the depths of the sulcus and extending anteriorly. The primary motor cortex corresponds to BA4. It includes the anterior bank of the central sulcus and much of the precentral gyrus (the

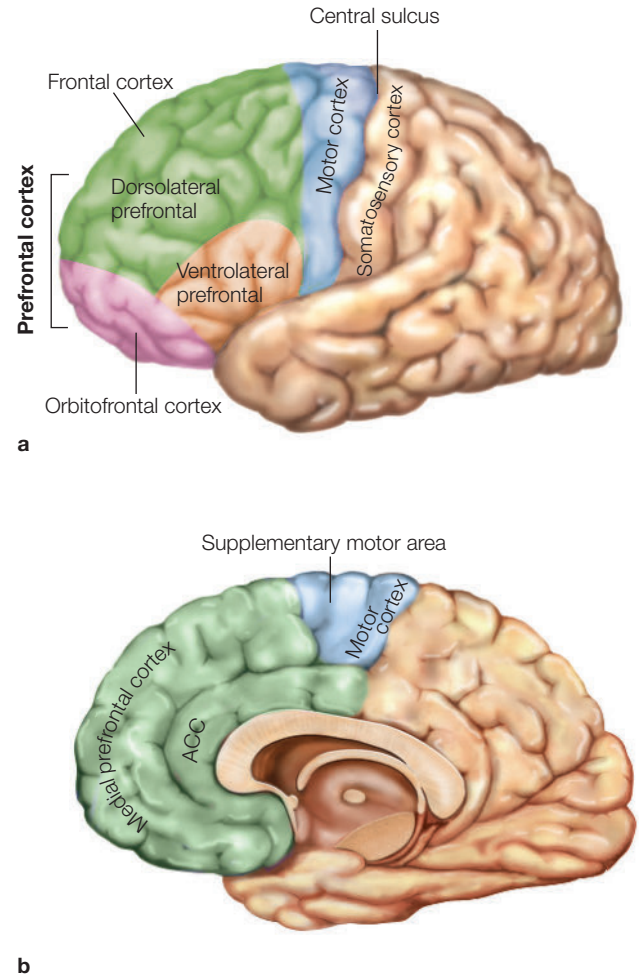


FIGURE 2.40 Cortical divisions of the frontal lobe.

(a) The frontal lobe contains both motor and higher-order association areas. For example, the prefrontal cortex is involved in executive functions, memory, decision making, and other processes.

(b) Midsagittal section of the brain, showing the medial prefrontal regions, which include the anterior cingulate cortex (ACC). Also visible is the supplementary motor area.

prefix *pre-* in neuroanatomy means “in front of”). It receives input from the cerebellum and basal ganglia via the thalamus and the premotor area. It is mainly responsible for generating neural signals that control movement. The output layer of primary motor cortex contains the largest neurons in the cerebral cortex: the pyramidal neurons known as *Betz's cells*. They reach 60 to 80 micrometers (μm) in diameter at the cell body, and some of them send axons several feet long down the spinal cord.

Anterior to the primary motor cortex are two motor association areas of cortex located within BA6: the *premotor cortex* on the lateral surface of the hemisphere, which contributes to motor control, and the *supplementary motor cortex*, which lies dorsal to the premotor area and extends around to the hemisphere's medial

surface, which is involved in the planning and sequencing of movement. Stimulating this latter region results in movement patterns that are much more intricate than those resulting from stimulation of the primary motor cortex. Motor association areas modulate initiation, inhibition, planning, and sensory guidance of motor movements, and they contain motor neurons whose axons extend to synapse on motor neurons in the spinal cord.

THE PARIETAL LOBE: SOMATOSENSORY AREAS What we know about the external world we learn through our senses. The parietal lobe receives sensory information about touch, pain, temperature sense, and limb proprioception (limb position) via receptor cells on the skin that convert it to neuronal impulses that are conducted to the spinal cord and then to the somatosensory relays of the thalamus (**Figure 2.41**). From the thalamus, inputs travel to the *primary somatosensory cortex (S1)*, a portion of the parietal lobe immediately caudal to the central sulcus. The next stop is the *secondary somatosensory cortex (S2)*, which is a unimodal association area that continues to process sensory information and is located ventrally to S1; S2 receives most of its input

from S1. Together, these cortical regions are known as the *somatosensory cortex*.

As sensory information moves from S1 to adjacent unimodal areas and then to multimodal areas, processing becomes increasingly complex. Lesions at various locations in the parietal lobe result in all sorts of odd deficits relating to sensation and spatial location: People with such lesions may think that parts of their body are not their own or that parts of space don't exist for them; they may recognize objects only from certain viewpoints, or be unable to locate objects in space at all. Stimulating certain regions of the parietal lobe causes people to have “out-of-body” experiences (Blanke et al., 2002; see Chapter 13).

Topographic mapping of the sensory and motor systems. The specific cortical regions of the somatosensory and motor cortices that process the sensations and motor control of specific parts of the body have been mapped out. The spatial relationships of the body are fairly well preserved in a topographic map of neural representations draped across these cortices (see **Box 2.2**). For example, within the somatosensory cortex, neurons that respond to touch of the index finger are adjacent to those that respond to touch of the middle finger, which are also next to neurons that respond to touch of the ring finger. Similarly, the hand area as a whole is adjacent to the lower arm area, which is near the upper arm, and so forth. There are discontinuities, however. For example, the genital region's representation is just below that of the feet.

This mapping of specific parts of the body to specific areas of the somatosensory cortex is known as **somatotopy**, resulting in somatotopic maps in the cortical areas. These maps are not set in stone, however, and do not necessarily have distinct borders. For example, the amount of cortical representation of a body part can decrease when that part is incapacitated and not used for extended periods, or it can increase with concentrated use. Loss of a limb can result in increased cortical representation of body parts represented adjacent to the representation of the lost limb. Topographic maps are a common feature of the nervous system (see Chapter 5).

THE OCCIPITAL LOBE: VISUAL PROCESSING AREAS

The business of the occipital lobe is vision. Visual information from the outside world is processed by multiple layers of cells in the retina and transmitted via the optic nerve to the lateral geniculate nucleus of the thalamus, and from there to V1—a pathway often referred to as the *retinogeniculostriate* or primary visual pathway

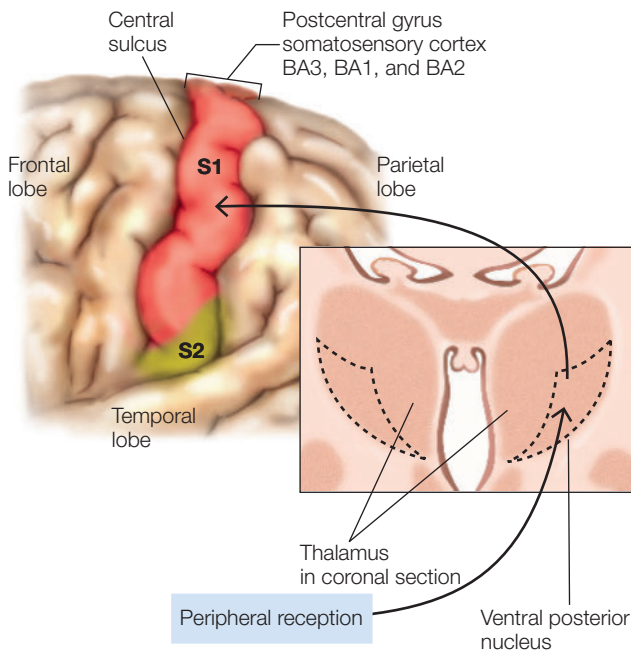


FIGURE 2.41 The somatosensory cortex, located in the postcentral gyrus.

Inputs from peripheral receptors project via the thalamus (shown in cross section) to the primary somatosensory cortex (S1). Secondary somatosensory cortex (S2) is also shown.

BOX 2.2 | LESSONS FROM THE CLINIC

Cortical Topography

From the pattern of muscle spasms that progress across the body during a seizure, John Hughlings Jackson, whom we met in the previous chapter, inferred that the cortical hemispheres contain an orderly, maplike representation of the body. Sixty years later, neurosurgeon Wilder Penfield and physiologist/neurologist Herbert Jasper (Jasper & Penfield, 1954), at the Montreal Neurological Institute, were able to directly stimulate the cortex of their patients undergoing brain surgery, and in doing so, they found evidence for Jackson's inference. Because there are no pain receptors in the central nervous system, brain surgery can be performed while patients are awake and able to verbally describe their subjective experiences. Penfield and Jasper, during the same procedure in which they were removing damaged brain tissue, systematically explored the effects of small levels of electrical current applied to the cortical surface.

They found a topographic correspondence between cortical regions and body surface with respect to

somatosensory and motor processes. This correspondence is represented in **Figure 2.42** by drawings of body parts being overlaid on drawings of coronal sections of the motor and somatosensory cortices. The resulting map of the body's surface on the cortex is sometimes called a *homunculus*, because it is an organized representation of the body across a given cortical area.

Note that there is an indirect relation between the actual size of a body part and its cortical representation: Body parts that have behaviorally relevant sensors or require fine motor movement have more cortical space dedicated to them on the sensory and motor maps. For example, areas within the motor homunculus that activate muscles in the fingers, mouth, and tongue are much larger than would be expected if the representation was proportional, indicating that large areas of cortex are involved in the fine coordination required when we manipulate objects or speak. Other animals also have topographically ordered sensory and motor representations. For example, rats have large areas dedicated to whisker sensations.

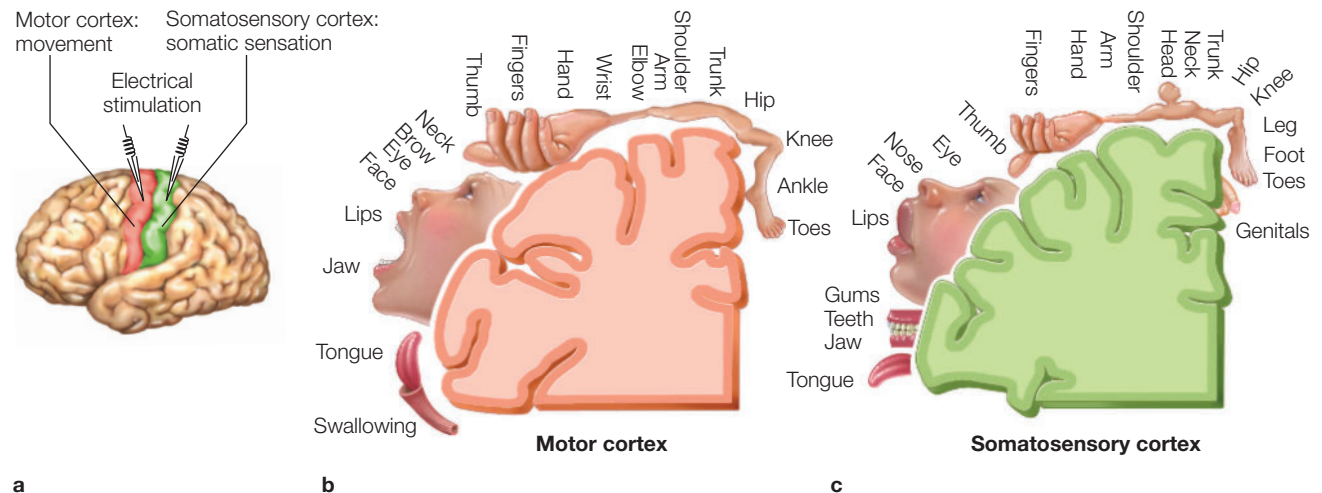


FIGURE 2.42 Topographic correspondence between cortical regions and body surface with respect to somatosensory and motor processes.

The two coronal sections, showing motor cortex (b) and somatosensory cortex (c), correspond to the colored regions highlighted in the lateral view of the whole brain (a). Only the left hemisphere is shown and the coronal sections are shown as viewed from a posterior viewpoint.

(**Figure 2.43**). The primary visual cortex is where the cerebral cortex begins to process visual information. As mentioned earlier, this area is also known as *striate cortex*, *V1* (for “visual area 1”), or *BA17*.

In humans, the primary visual cortex is on the medial surface of the cerebral hemispheres, extending only

slightly onto the posterior hemispheric pole. Thus, most of the primary visual cortex is effectively hidden from view, between the two hemispheres. The cortex in this area has six layers and begins the cortical coding of visual features like luminance, spatial frequency, orientation, and motion—features that we will take up in detail in

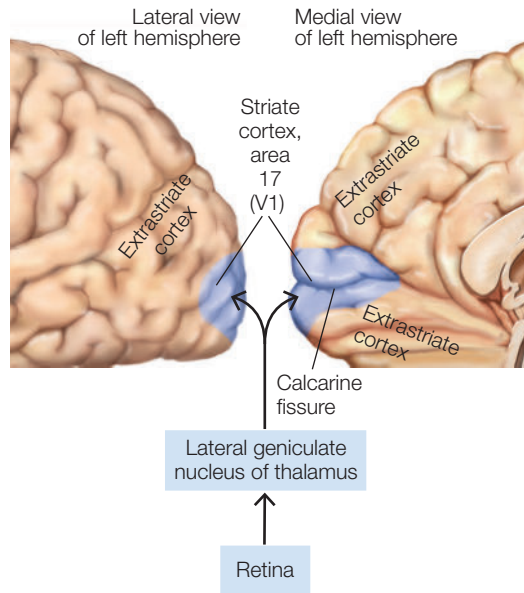


FIGURE 2.43 The visual cortex, located in the occipital lobe. Brodmann area 17—also called the *primary visual cortex*, visual area 1 (V1), and striate cortex—is located at the occipital pole and extends onto the medial surface of the hemisphere, where it is largely buried within the calcarine fissure.

Chapters 5 and 6. Surrounding the striate cortex is a large visual unimodal association area called the *extrastriate* (“outside the striate”) visual cortex (sometimes referred to as the *prestriate* cortex in monkeys, to signify that it is anatomically anterior to the striate cortex). The extrastriate cortex includes BA18, BA19, and other areas.

The retina also sends projections to other subcortical brain regions by way of secondary projection systems. The superior colliculus of the midbrain is the main target of the secondary pathway and participates in visuomotor functions such as eye movements. In Chapter 7 we will review the role of the cortical and subcortical projection pathways in visual attention.

THE TEMPORAL LOBE: AUDITORY PROCESSING AREAS

Neural projections from the cochlea (the auditory sensory organ in the inner ear) proceed through the subcortical relays to the medial geniculate nucleus of the thalamus and then to the primary auditory cortex. The primary auditory cortex lies in the superior temporal gyrus of the temporal lobe and extends into the lateral sulcus, where the transverse temporal gyri, which make up a region known as *Heschl's gyrus* that roughly corresponds with Brodmann area 41, are buried (**Figure 2.44a**). Previously, the auditory cortex was subdivided into primary auditory cortex (A1), secondary auditory cortex (A2), and other association areas. These are now referred to, respectively, as the core (BA41), belt (BA42), and parabelt areas (**Figure 2.44b**). The

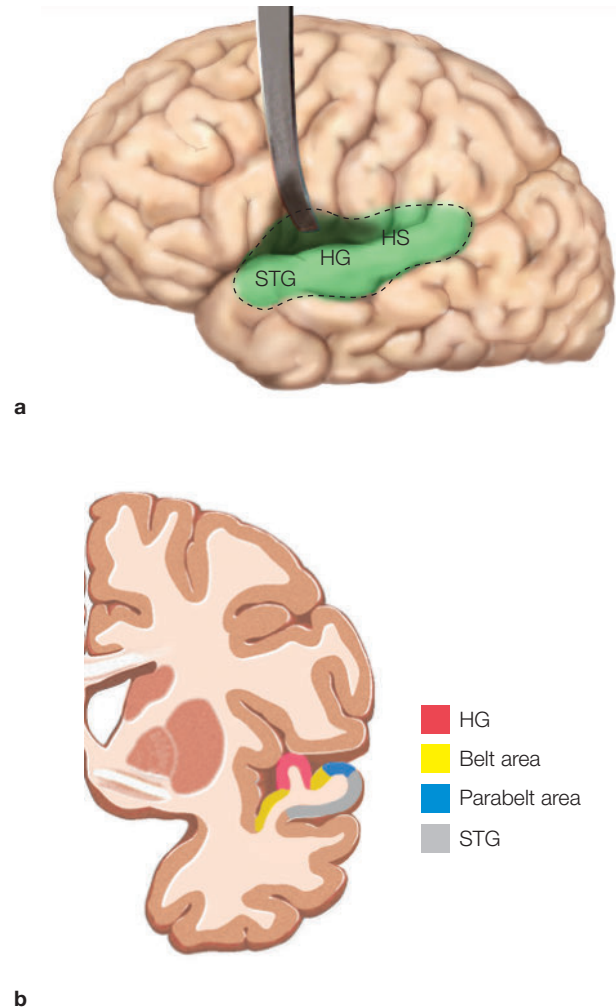


FIGURE 2.44 The auditory cortex. (a) Primary auditory cortex is located within the transverse temporal gyri (Heschl's gyrus, HG) buried within the lateral sulcus located in Brodmann area 41 of the temporal lobe. The primary auditory cortex and surrounding association auditory areas contain representations of auditory stimuli. HS = Heschl's sulcus; STG = superior temporal gyrus. (b) Coronal view, showing Heschl's gyrus within the lateral sulcus.

parabelt area surrounds the auditory cortex and aids in the perception of auditory inputs; when this area is stimulated, sensations of sound are produced in humans.

The auditory cortex has a tonotopic organization, meaning that the physical layout of the neurons is based on the frequency of sound. Neurons in the auditory cortex that respond best to low frequencies are at one end, and those that respond to high frequencies are at the other. The macroanatomy, microanatomy, and tonotopic pattern vary substantially across individuals. While sound localization appears to be processed in the brainstem, processing in the auditory cortex is likely essential for performing more complex functions, such as sound identification. Its role in hearing remains unclear and under investigation.

ASSOCIATION CORTEX If someone were to make you angry, would you have the “knee-jerk” reaction of punching that person? As mentioned earlier, the knee jerk is a result of a monosynaptic, stimulus–response reflex arc that leaves cortical processing out of the loop (Figure 2.25). Most of our neurons, however, feed through a series of cortical synapses before the behavioral response occurs. These synapses provide integrative and modular processing in association areas, the outcomes of which are our cognition—including memory, attention, planning, and so forth—and our behavior, which might or might not be a punch.

As mentioned earlier, a good portion of the neocortex that is neither primary sensory cortex nor primary motor cortex has traditionally been termed **association cortex**. Each primary sensory area is linked to its own unimodal association area that receives and processes information from only that sense. For example, though the primary visual cortex is necessary for normal vision, neither it nor the extrastriate cortex is the sole locus of visual perception. Regions of visual association cortex in the parietal and temporal lobes process information from the primary visual cortex about color, simple boundaries, and contours to enable recognition of these features as, for example, a face. Moreover, visual association cortex can be activated during mental imagery when we call up a visual memory, even in the absence of visual stimulation.

Unimodal areas are not connected to each other, protecting the sensory reliability of experiences and delaying cross contamination from other senses (Mesulam, 2000). While unimodal areas can encode and store perceptual information, and while they can identify whether sensory features are the same—say, comparing two faces—without information from other modalities, they don’t have the ability to tie that perception to any other experience. Thus, within a unimodal area the face remains generic; it can’t be recognized as a specific person, and no name is attached to it. Those tasks require more input, and to get it, processing moves on to multimodal cortex.

Multimodal association cortex contains cells that may be activated by more than one sensory modality. It receives and integrates inputs from many cortical areas. For example, inputs of the various qualities of a particular stimulus (e.g., the pitch, loudness, and timbre of a voice) are integrated with other sensory inputs—such as vision, along with memory, attention, emotion, and so forth—to produce our experience of the world. Multimodal areas are also responsible for all of our high-end human abilities, such as language, abstract thinking, and designing a Maserati.

THE FRONTAL LOBE REVISITED: PREFRONTAL CORTEX

The more anterior region of the frontal lobe, the **prefrontal cortex**, is the last to develop and is the evolutionarily youngest region of the brain. It is proportionately larger in humans compared to other primates. Its main regions are the dorsolateral prefrontal cortex, the ventrolateral prefrontal cortex, the orbitofrontal cortex (Figure 2.40a), and the medial prefrontal regions, including the anterior cingulate cortex (Figure 2.40b).

The prefrontal cortex takes part in the more complex aspects of planning, organizing, controlling, and executing behavior—tasks that require the integration of information over time. Because of its facility with these tasks, the frontal lobe is often said to be the center of cognitive control, frequently referred to as *executive function*. People with frontal lobe lesions often have difficulty reaching a goal. They may know the steps that are necessary to attain it but just can’t figure out how to put them together. Another problem associated with frontal lobe lesions is a lack of motivation to initiate action, to modulate it, or to stop it once it is happening. The frontal lobes also are involved in social functioning, so inappropriate social behavior often accompanies frontal lobe lesions.

PARALIMBIC AREAS The paralimbic areas form a belt around the basal and medial aspects of the cerebral hemispheres and don’t reside in a single lobe. They can be subdivided into the olfactocentric and hippocampocentric formations. The former includes the temporal pole, insula, and posterior orbitofrontal cortex; the latter includes the parahippocampal cortex, the retrosplenial area, the cingulate gyrus, and the subcallosal regions. The paralimbic areas lie intermediate between the limbic areas and the cortex, and they are major players linking visceral and emotional states with cognition. Processing in these areas provides critical information about the relevance of a stimulus for behavior, rather than just its physical characteristics, which are provided by the sensory areas. This information affects various aspects of behavior reviewed in our earlier discussion of the limbic system.

LYMBIC AREAS While most of the structures in the limbic system are subcortical, the limbic areas of cortex lie on the ventral and medial surfaces of the cerebral hemispheres and include the amygdala, the piriform cortex, the septal area, the substantia innominata, and the hippocampal formation. These areas have the greatest number of reciprocal interconnections with the hypothalamus. Like the paralimbic areas, the limbic areas are major contributors to behavior via the roles they play in processing emotion, arousal, motivation, and memory.